WELCOME BOOKLET FOR NEW ICUBE PhD STUDENTS

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1 Introduction

« What do we mean by someone who does well in college? Someone who does well in their doctorate? A good doctoral student, I think, manages to do something other than their

doctorate, manages to work effectively when they are at work but manages to have a life outside of the doctorate. And that, I think, is the most difficult thing to achieve.»
Conference Sit down and write Your Dissertation! Practical tips for scientific writing (University of Strasbourg, 2015) Geneviève Belleville, professor at Laval University, Quebec

Welcome! After consulting your file and interviewing you, your supervisors have chosen you among other candidates to work with them. Congratulations and welcome among us! You are a new PhD student at the ICube Laboratory. Whether or not you have already studied your future field of research, these first years of research will give you the opportunity to put to profit your knowledge and know-how acquired during your studies and will also test your perseverance, your rigor and your ability to clearly present your work. At the end of these years of thesis, you will have become a specialist on your subject of study and through your work, you will have advanced your field. The experience and knowledge will allow you to pursue your career, in academic research, teaching or corporate research and development with ease.

What this booklet does and does not contain This welcome booklet is here to introduce you to the essential elements of a thesis, the functioning of the laboratory and guide you through the different steps. Thus armed, we hope that you will quickly get your bearings within the laboratory. The goal here is not to detail all the procedures that exist, but rather to show you where to find information and who to turn to in different situations. A booklet already exists on the steps to take in daily life. It was written by the association ADDAL.

Who are we ? The ICube Young Researcher Association (Association des Jeunes Chercheurs d'ICube (AJCI)) was founded in 2019 and aims to animate the social and scientific life of ICube's PhD students and postdocs by organizing seminars and convivial events. If you want to join us and help us in the organization of events, you can send an email to ajci@icube.unistra.fr . A discord server is also available to facilitate exchanges between PhD students: https://discord.gg/5CTdudvJd9 .

2. Registration procedures

The registration and re-registration procedures, both pedagogical (at the doctoral school) and administrative (at the University of Strasbourg) are fully described on the MSII doctoral school website, here .

As soon as your registration is finalized, you will have to activate your Digital and Social Work Environment (Ernest platform), on which you will be able to find your schooling certificate for the current academic year, and access your e-mail. As a student, you are entitled to a student card that will be sent to you by mail, which offers you many advantageous discounts. As a member of the university of Strasbourg, you also have access to all the sports activities offered by the university (see the suaps website)

3 Life in the laboratory

3.1. Intranet

ICube has an intranet and a wiki (sub-part of the intranet), which contain a large number of internal documents. Once you have registered for your thesis, it is essential that you make your first connection to the intranet, otherwise you will not be considered as a member of ICube.
You will be able to reserve resources such as meeting rooms, software tokens, edit your office label, access mailing lists. You will also find many documents that will allow you to know the laboratory better (especially the internal rules).

3.2. Procedures for newcomers

In order to be considered a member of the laboratory, it is mandatory to complete the procedure for newcomers .

3.3 Presentation of team and sites

The ICube laboratory is located on 5 different sites. This map shows the geographical location of the different sites, as well as the 17 teams and technological platforms present on each of them.

3.4. Attached doctoral schools

Two doctoral schools are attached to ICube laboratory

- ED269: Doctoral school of Mathematics, Information and Engineering Sciences (MSII). This doctoral school brings together the majority of ICube's doctoral students

- ED414: Doctoral School of Life and Health Sciences

3.5. Events for doctoral students

3.5.1 MSII poster day

The poster day is an event for the doctoral students of the MSII doctoral school and takes place in early October. The director of the doctoral school presents the functioning of the doctoral school and the training obligations to newcomers. The PhD students starting their second year present their work in the form of a poster. The best poster is selected and wins a prize.

3.5.2 Welcome day for newcomers

The welcome day for newcomers is organized in winter, this day is intended for all newcomers (not only PhD students and postdocs). The morning starts with a welcome

speech by the laboratory director, followed by a presentation of the laboratory. After the lunch buffet, you will have the choice to visit one of the 5 Cube sites.

3.5.3. Information meeting on research careers

The information meeting on research careers presents the pursuit of a career in research after the doctorate. It presents the different job opportunities as well as the steps (qualification and recruitment competition) to enable you to obtain a position as a researcher, teacher-researcher, research engineer, etc. Members of the laboratory are part of the CNU sections managing the qualification (to contact one of these members, please contact Anne-Sophie Goudot (asgoudot@unistra.fr)). The most recent editions also include a presentation on research in companies and the advantages you can bring to the table during recruitment. This meeting is important, as it allows you to build a strategy to prepare for the rest of your thesis and can guide your choices of training, teaching and participation in events throughout your thesis. To contact a member of a qualification jury within the laboratory, please contact Anne-Sophie Goudot (asgoudot@unistra.fr).

3.5.4. Inter-teams olympiads and other social events

Every year, ICUBE organizes an inter-team Olympiad during the summer. This takes the form of three friendly events organized (one per month) during which ICUBE teams compete on funny, intellectual and funny activities (basketball, board games, molkky, petanque, chess ...). The team that finishes first at the end of the three events leaves with a magnificent TOTEM, which is awarded every year.

Other social events for PhD students are organized during the year, and will be announced by email, so watch out for emails starting with [AJCI] !

3.6 Representation of doctoral students in the laboratory

3.6.1. Doctoral students' council

The doctoral students council brings together two to three times a year the elected representatives of each team with the laboratory director. This is the main forum for dialogue between the laboratory director and the doctoral students. Doctoral students can bring up their difficulties, give their feedback on the different events, and suggest improvements.

3.6.2. The council of the laboratory

The council of the laboratory is composed of the deputy directors and the elected representatives of the personnel, including three doctoral students who are part of the doctoral students' council. It gives its opinion on decisions and orientations of the laboratory (research program, recruitment, budget, ...).

3.7. Professional card

As a doctoral student, you are a staff member of the laboratory and you have the right to a professional card that allows you, among other things, to eat on the staff side. Go to this page to edit your card: https://www.passcampus.fr/professionnel/ .

3.8. ICube publications

On this page of the intranet, all publications of ICube members are referenced. During your thesis, you will publish scientific articles. If they are accepted, please add them to the database.

3.9 Missions

During your thesis, you will probably go on a mission (for example to participate in a scientific conference). Details on the procedures to follow to go on a mission are available here.

3.10. Reimbursement of public transport and Sustainable Mobility Package

ICube reimburses part of the cost of public transportation. If you wish to obtain the compensation, you can contact Emilie Acédo (erioux@unistra.fr), who will send you the form to fill out.

The University of Strasbourg has set up a Forfait Mobilité Durable (sustainable mobility package) in order to encourage ecological means of transportation (bicycle, carpooling). It offers a compensation of 200 euros per year.

4 Progress and follow-up of the thesis

4.1 Entry into the research world

4.1.1. The importance of publishing

You are about to enter the world of research, and in order to make your way in it, it is essential to start publishing during the PhD. Indeed, a researcher's reputation is largely determined by his or her publications, and more importantly, it is almost impossible to get the qualification (see section 6.3) without publication. There are two categories: for a scientific journal or for a conference. Conference papers are usually short (4 pages), and allow you, if the paper is accepted, to present your research at the conference in question. Journal papers are longer (a dozen pages), which allows you to go into more detail. In both cases, the paper will be submitted to a peer review system. A review committee will evaluate your

submission and decide whether to accept your paper directly, request minor or major changes, or reject it.

It is important to choose the right journal (or conference) to submit your paper to. It should be neither too selective (because you risk being rejected), nor too little because a publication in a good scientific journal is an asset on your CV. It is therefore important to discuss with your supervisors the most appropriate journal. To give you an idea of the value of a scientific journal, there is a metric: the impact factor. This metric is much debated, because it is imperfect. Thus, many organizations have now signed Dora, the San Francisco Declaration on Research Evaluation, which revisits this notion. Once again, the most important thing is to discuss your publication target with your supervisors who know the publications and their community. You will find here a (non exhaustive) list of scientific journals.

4.1.2. Access the publications of other researchers

When an article is published in a scientific journal, it is not freely available. To allow researchers and students to access scientific articles, the University of Strasbourg has paid the access rights for many articles. As a member of the University of Strasbourg, you can therefore access them. To do so, you just have to add ".scd-rproxy.u-strasbg.fr" after the address of the journal, and to enter your unistra identifiers. For example replace https://ieeexplore.ieee.org/document/9224166 by https://ieeexplore.ieee.org.scd-rproxy.u-strasbg.fr/document/9224166 . You can also install the EZProxy extension which handles this automatically.

Many authors also deposit their publications on arxiv, an open archive for electronic pre-publication, in order to make their articles freely available. When you consult an article on arxiv, it is important to check if it has been accepted by a peer-reviewed journal, to ensure its scientific validity. Finally, it is also important to know, as a future publisher, that the Digital Republic Act of 2016 allows an author to deposit in an open archive the version accepted for publication of a journal article even if he or she has assigned all rights to a publisher. An embargo (blocked period) cannot exceed 6 months in science, technology and medicine. Beyond that, it is possible for each French researcher to put online the version accepted for publication (the final text, before the layout by the editor). To know the rules for each journal, Sherpa/Romeo is available.

4.1.3. Research groups

In France, there are many research groups (GdR) associated with various themes. The mission of a research group is to animate the French scientific community in the field in question, and to structure research activities. It is important to identify the research groups in your field in order to be aware of the conferences, summer or winter schools and other scientific events organized by the research group. It is often an opportunity to make yourself known, to exchange with other specialties of your subject, ask your supervisors to participate in these events!

4.2 Committee of follow-up

The Committee of follow-up is an essential step in the doctoral process. The committee is composed of at least two members, including an expert in the field of the thesis and an external member not affiliated with the laboratory. One of the

members must hold an HDR (Habilitation à Diriger des Recherches) or its equivalent abroad, and ideally, a member from outside the institution of enrollment.

The committee takes place during the first year of the thesis, preferably in the second half of the year. It is mandatory for enrollment in the following academic years and must be declared to the doctoral school in the second semester of the first year of the thesis. The meeting can be held in person or virtually, with a deadline usually set for September 15th, the deadline for thesis pre-registration.

The objectives of the monitoring committee include a mid-term evaluation of the thesis work, providing the doctoral student with critical reflections and advice on various aspects such as the methodological approach and results obtained. It also aims to assess the progress of the thesis, discuss the details of the defense or the funding for an additional year if necessary, and prepare for post-thesis activities, including potential post-doctoral positions.

The committee meeting does not necessarily require a scientific presentation, but it should include a discussion time with the doctoral student, a time with the committee excluding the student, and a discussion involving all participants. It is imperative to fill out the required documents. The monitoring committee is considered a collaborative step aimed at supporting the doctoral student in their doctoral journey, rather than a stressful examination.

4.3 Disciplinary and transversal courses

The decree of August 7, 2006 (Légifrance) requires each doctoral school to manage the training of its doctoral students. No matter which doctoral school you belong to, you are concerned, from the first to the last year of your thesis.

Since each course and each student is unique, it is up to you to create your own training plan. An exchange with your thesis director is welcome. This is called the Individual Training Plan (PIF). The objective of the PIF is to encourage you to think about your research and your future career by allowing you to identify the training that would be useful to you. The PIF will evolve with your thesis and your needs and desires. The training courses in which you wish to participate must be indicated in the training agreement (see the documents to be provided during registration).

4.3.1 Disciplinary courses

The disciplinary training (scientific training) is composed of two parts: scientific modules and seminars.

The scientific modules are the classic modules associated with the University's Master's programs. You must choose at least two different modules over the duration of your thesis for a quota of approximately 40 to 50 hours. There is no training catalog provided by the ED. On the other hand, you can find on this page a list of the modules proposed by the University of Strasbourg : https://en.unistra.fr/study/find-a-course .

You are also obliged to attend 6 seminars per thesis year, at least 4 of which are organized by your ED. Seminars are regularly announced by mail with all the modalities of participation.

4.3.2 Transversal courses

The cross-disciplinary training courses allow you to acquire the notions of the cross-disciplinary competences for your thesis project. 3 cross-disciplinary courses are compulsory: "Integrity charter in Scientific profession", and "MOOC Research integrity in scientific professions" in the first year and "Scientific Communication and Research Culture" in the second year. In addition to these 3 courses, you must take one or more other modules of your choice, for a total of 7 to 12 hours. You will find a catalog of the cross-disciplinary courses offered on the AMETHIS website.

There are other organizations whose courses are not listed in AMETHIS! Among these organizations, we can mention the Urfist, 'Institut de Développement et d'Innovation Pédagogiques' (IDIP) or Jardin des Sciences.

4.3.3 Registration of courses

To validate the training courses you have done, you must register on AMETHIS. For the scientific seminars and the transversal trainings, you must register beforehand on the site. You will find the agenda of the seminars and trainings in the tab 'sessions calendar', then you just have to click on the desired training then on 'apply to this session'.

For the scientific training courses, you must have a document signed by the person in charge of the module justifying the hours worked. This document is available on the ED MSII website, tab 'disciplinary training'. Then you can register the training on AMETHIS. To do so, go to the tab 'equivalences catalog', then click on the '+' on the top left.

4.3.4 Useful links

Here are some useful links where you can find additional information.
ED MSII - doctoral courses : link
ED MSII - disciplinary training : link
ED MSII - transversal trainings : link
IDIP : link
Jardin des Sciences : link
Program of the Urfist of Strasbourg : link
Unistra's Master's program : link

Amethis : Amethis (unistra.fr)

4.4 Rédaction

During the last year of the PhD, every student is required to write a thesis manuscript, putting on paper his or her research process and results. We cannot give a generic plan applicable to all, but there are some remarks we can make and some information to keep in mind.
The manuscript is a highly technical and structured document. Care must be taken to define all terms used and everything must be referenced. Some constraints are necessary. The University of Strasbourg provides a front and back cover that must be used for the manuscript. As for the format (Word, LaTeX...), you must ask your thesis director directly. This also applies to the citation standard. The University also offers a Word style sheet ready to be used. There are some tools that can facilitate the writing work, for example Zotero to manage references and citation styles.
There are a certain number of legal obligations concerning the thesis manuscript, be it towards the doctoral school, the University, the constitution of the thesis jury, the deposit of the manuscript on the thesis database... We will not detail these obligations here. There is an FAQ concerning these points on the University's website.
We strongly encourage you to be aware of these constraints early on: some of them should (ideally) be done before writing (e.g., ask for permission to write in English)
Unistra - Conseils techniques : lien
Unistra - Soutenance et obligations : lien
ED MSII - soutenance : lien
Bibliothèque Universitaire - FAQ thèse : lien

4.5 Defense

Once the manuscript is written, it takes at least two months before it can be defended. Indeed, the manuscript has to be read again by referees and various corrections have to be made. During this period, the preparation of the defense really begins. The first step is to choose the jury. It is composed of 4 to 8 people including the referees and is proposed to the doctoral school by the thesis director. For your part, you will have to send a summary of your thesis to the ED. At least 3 weeks before the defense, you will have to inform the ED of the planned date of the defense. Once the rapporteurs return their report, the defense can take place.
The defense is the moment when the whole thesis work is presented. At the end of the defense, the title of doctor will be conferred. It is also a kind of ceremony. The steps are systematically the same. The members of the jury are introduced by the president. The doctoral student then presents his/her work. The president of the jury then gives the floor to the rapporteur to ask questions and then to the other members of the jury. He closes the questions and invites the doctoral candidate to leave in order to proceed with the deliberation. The candidate is then called back and the president summarizes the discussion that took place. He then gives the title of doctor. The defense is then officially over, very generally a drink is organized by the doctor to celebrate the event.

Once the defense is over, there are still some documents to be signed and sent to the ED in order to make the diploma official. They will not be detailed here.
For more information, you can read the section Soutenance of the ED 269 website.

5. Complementary missions

5.1 Teaching missions

It is possible, in parallel with your PhD, to teach in one of the University of Strasbourg's departments or in the ICube laboratory's supervisory institutions. If you plan to become a lecturer after your thesis, it is important to have taught during your doctorate (ideally, at least 100 hours). However, you should discuss this with your director. A teaching assignment is a non-negligible workload that can (or will) slow down your research work.

In most cases, doctoral students are responsible for teaching TDs, TPs and project supervision. In exceptional cases, doctoral students may be given a lecture.

The IDIP offers training in pedagogy, which can be useful for developing your teaching methods. Moreover, this training is very valuable if you wish to apply for a position as a lecturer.

There are two types of status: you can be a temporary lecturer (‘vacataire) or a teaching assistant (‘missionné d’enseignement’):
- A teaching assistant must ensure 64 hours of teaching. Not one more and not one less. Moreover, he is paid every month.
- A temporary teacher can work much less (a few hours) or much more (up to 96 hours per year). This is a much more flexible status, but they are paid only once at the end of the semester (or even at the end of the year). The salary depends on the number of hours worked.

After you register for your thesis, you will receive a form from the doctoral school, which will allow you to apply to the component of your choice. You will find below specific information for each component.

5.1.1 IUT Robert Schumann

The IUT Robert Schumann includes 5 departments: Chemistry, Civil Engineering, Information-Communication, Computer Science, Marketing Techniques. The diplomas prepared are the BUT and Professional Licences. The training courses are from bac+1 to bac+3 level.

To obtain information about the courses, you can contact the direction of studies:
diret-in@iutrs.unistra.fr

5.1.2. Haguenau IUT

The Haguenau IUT includes 3 departments: MMI (Multi-Media and Internet), GEII (Electrical and Computer Industrial Engineering) and QLIO (Quality, Industrial Logistics and Organization). The diplomas prepared are the BUT and Professional Licenses. The training courses are from bac+1 to bac+3 level. The courses are very varied and include computer science/programming, web, 3D, industrial computer science, electronics, automation/automatics, industrial engineering, industrial production management, logistics, quality and all other subjects related to the BUT courses at the IUT. You will find on this link a presentation of the courses.

To obtain information about the courses, you can contact the department heads and their directors of studies (iuthag-direction@unistra.fr ou iuthag-directoire@unistra.fr).

5.1.3 Louis Pasteur IUT

The IUT Louis Pasteur groups together 4 departments : Biological Engineering (GB), Industrial Engineering and Maintenance (GIM), Physical Measurements (MP), Business and Administrative Management (GEA)

The courses are at the bac+3 level in a wide variety of fields (Mathematics, Physics, Computer Science, Statistics, Quality/Safety, Environment, Management, Labour Law, Sustainable Development, Biology, Physiology, Microbiology, Immunology, Chemistry, Biochemistry and many others)

To obtain lessons, it is necessary to contact according to the field concerned:

- B.U.T. Biological Engineering (GB) --> iutlps-gb@unistra.fr
- B.U.T. Industrial Engineering and Maintenance (GIM) --> iutlps-gim@unistra.fr
- B.U.T. Physical measures (MP) --> iutlps-mp@unistra.fr
- B.U.T. Business and Administration Management (GEA) --> iutlps-gea@unistra.fr

5.1.4 The UFR of Mathematics and Computer Science

The UFR of Mathematics and Computer Science is composed of two departments
- the department of mathematics (72 permanent staff) and
- the department of computer science (35 permanent staff)

The first semester is common to both mathematics and computer science. From the second semester onwards, students choose their path and continue in the computer science or mathematics degree. In computer science, one can stop at the licence level by choosing a professional licence (ARS), but one can also go further on to a master's degree. Each

department has several master's courses that allow students to specialize in pure mathematics, applied mathematics, statistics, magistracy, and preparation for the agrégation. In computer science, students can specialize in complex systems and artificial intelligence, software engineering, imaging and 3D, or systems and networks.

The subjects taught are described on their website in the menu Formations. For most of the courses, there is a written support which allows the students (but also the teachers who discover the subject) to prepare the work. TDs and TPs are sometimes accompanied by answers, but not always.

In the computer science department, in August or September, a meeting is organized where the directors of studies present the department to new teachers.

To obtain lessons you can contact
Jean-Luc Dortet-Bernadet (dortet@math.unistra.fr) for the mathematics department
Eric Violard (violard@unistra.fr) for the computer science department

5.1.5 Télécom Physique Strasbourg

You will find a presentation of the school here . The subjects taught are at the engineering level (bac +3 to bac +5), and M1/M2. A presentation of the training is available here. The profile of the doctorate sought to teach corresponds to that of sections CNU 27, 28, 61 and 63 (more needs in 61, 63 and 28). In addition to the 64 hours of teaching, you are required to supervise at least 3 exams.

To obtain teaching positions, you can contact the director of studies Vincent Mazet (vincent.mazet@unistra.fr), who will interface with the teaching teams.

5.1.6 Faculty of Physics and Engineering

The level goes from Licence 1 to Master 2. They are described on their website. To apply to the faculty of physics and engineering you can contact the director of studies Herve Berviller (herve.berviller@unistra.fr)

5.1.7. Faculté de Géographie et d'aménagement

The Faculty of Geography offers 4 main degrees: 1 bachelor's degree in geography and planning and 3 master's degrees in geography including a master's degree in Earth Observation and Geomatics. Geography is the discipline at the heart of the courses, but the faculty is multidisciplinary. Social sciences, earth sciences, and computer science are disciplines that are also addressed at different levels. A doctoral student who does not have a geography degree can therefore also teach there. Applications and selections are made in June.

If you are interested in teaching at the Faculty of Geography, you can contact Damien Ertlen (damien.ertlen@live-cnrs.unistra.fr)

5.2. Expertise mission carried out in a company, a local authority or an administration

The expertise mission is an agreement established between a company and the university where a doctoral student can carry out consulting activities (doctoral consultant) for a maximum duration of 32 working days. The 32 days of missions can be spread out throughout the academic year, and the additional salary for the doctoral student is smoothed out and paid throughout the academic year, just like for the other missions. The establishment of the agreement requires the agreement of the thesis director as well as the representative of the interested company.

If you wish to obtain more information and establish an agreement, you can contact Joëlle Hube (joelle.hube@unistra.fr)

5.3. Other complementary missions

There are two other types of complementary missions: Mission of dissemination of scientific and technical information and mission of valorization of scientific and technical research results. Unfortunately, we do not have any information about them

6. After the PhD

What to do after a PhD? This is a question that you should start thinking about at the beginning of your third year of thesis. The first step is to focus on what you want to do. Do you like teaching, research only or both? Are you more interested in the public or private sector? Are you interested in project management? Is finding a permanent post after your thesis a prerequisite? Is money your main motivation? By answering these questions, ideas about your future may come to you. The second step is to find the type of job that would suit you. A list of the main research jobs in France, each with a description, is given below to guide you in your reflection. The third step will be up to you to find and apply for jobs. You can also make an appointment at the Espace Avenir of the University of Strasbourg to discuss with experts in job search and career development. Unistra also offers the possibility of associating you with a mentor, who can help you in your professional transition. There are other organizations that help doctoral students in their post-doctoral career pursuits:

- the MOOC PhDOOC
- The New Thesis Chapter: NPC
- The Association Bernard Gregory : ABG , especially for the insertion in the private sector
- The PhD Talent Career Fair, an annual networking event for doctoral students wishing to enter the private sector after their thesis

6.1 Post-doc

Research / Teaching: *80 % / 20 %*
Sector : *public or private*
Duration : *≈ 6 months to 2 years*
Salary : *highly variable according to contract*

What is a Post-Doc ?
The post-doc is a post-doctoral research work, generally in the continuity of the work done during the PhD. It is conducted in a French or foreign host laboratory. Its duration is often one year, but several post-docs can be done after a PhD. It represents a considerable asset for obtaining a position as a lecturer or research fellow at the CNRS, especially if it is carried out in a different laboratory from the one where the doctorate was obtained. Recruiters appreciate young, mobile researchers because they are adaptable.

How to get a Post-doc ?
To obtain a post-doc, you must find a host laboratory and funding. The search for a host laboratory is generally done through the contacts of your thesis director. If your thesis is part of a national, European or even international project, the project partners are additional contacts for finding a laboratory. It is also important to make contacts during conferences and to be aware of the different laboratories in the world working on a subject similar to yours. Concerning the financing, some host laboratories take care of their post-doc. The post-doc can also be integrated in projects financed by the ANR. If this is not the case, you should contact other external organizations offering post-doc funding grants.

Some offers are posted on community mailing lists (for example robotics worldwide, imageworld)

6.2 ATER

Research / Teaching: *full ATER : 50%/50% , Half-ATER 75%/25%*
Sector : *public*
Duration : *≈ 1 year*
Salary : *available* **here**

ATERs, for Attachés Temporaires d'Enseignement et de Recherche (Temporary Teaching and Research Associates), have the status of contractual teacher-researcher. By equivalence, it can be situated between the monitorat and the position of lecturer. This one-year contract often constitutes a possibility of financing in case of need to extend one's thesis beyond three years. It is also a temporary alternative while waiting for an appointment as a lecturer, and an additional experience for the application to this position.

A full-time ATER must provide a full teaching service of 192 ETD hours per year (TD equivalent hours). A half-time ATER is required to provide half service of 96 FTE hours per year. A half-time ATER position frees up more time for research, but the salary is also strongly impacted. No additional teaching load can be assigned to them. In both cases, the

ATER participates in the various obligations involved in their teaching activity: supervision of students, knowledge control and examinations.

Although the required service is 192 hours ETD per year, it is interesting to know that some universities (including the University of Strasbourg) consider one hour of TP equivalent to one hour of TD for ATERs (an equality that is also valid for doctoral students on a teaching mission).

How to become ATER ?

The Galaxie portal for higher education personnel of the Ministry of Higher Education and Research theoretically lists all ATER positions in France to be filled for the coming year. However, some information is not always available on the site, so it is preferable to monitor in parallel the job offers on the sites of the different universities. The publication campaign for ATER offers is very staggered, with positions generally appearing between late February and early July.

For each online offer, candidates can submit an application file on the Galaxie portal. Paper applications can also be requested by the institutions. Applications that are declared admissible are examined by a committee of specialists from the institution, who propose the successful candidates. Few universities hold auditions, as the application file is often the only determining factor. The Curriculum Vitae (CV) and the letter of motivation must therefore be given great attention.

6.3 Lecturer

Research / Teaching: *80 % / 20 %*
Sector : *public*
Duration : *CDI (*Open-ended contract*)*
Salary : *The salary depends on the seniority and evolves according to the grid available* here

A Lecturer (MCF for 'Maitre de conférence') hired right after his PhD starts at step 2, if he has done post-docs, he can start at a higher step. (seniority is counted from the beginning of the PhD).

What is a lecturer ?
A lecturer is a teacher-researcher who holds a position in a public institution of higher education and research. He or she has the dual mission of ensuring the development of fundamental and applied research and transmitting the resulting knowledge to students. Their role is therefore multiple and full of responsibilities. He/she must provide an annual teaching service of 192 hours ETD. He/she supervises students in internships or thesis. He/she seeks local, national and international partnerships as well as funding. And he/she also produces scientific research.

Obtaining a qualification as a lecturer after a PhD is easy and quick, but to be recruited you have to be a little patient because the competition is tough on the job market. In the field of Technical Sciences, the average age of recruitment of lecturers is 32 years (source). This is only an average; it is quite possible, with a good record, to be recruited 1 year after one's doctorate. Hence the importance of teaching and training in pedagogy (at

the IDIP for example) during one's doctorate. Responsibilities in doctoral student associations, or participation in the organization of scientific events (e.g. science festival) are also a plus.

How to become a Lecturer ?

Upon obtaining the doctorate, the student must first be qualified to become a lecturer by a section of the Conseil National des Universités (CNU) (National Council of Universities). Once the qualification is obtained, he/she can apply for the recruitment competitions open to each post in each public institution of higher education and research. All the information concerning the qualification and the positions to be filled is given on the Galaxie portal for higher education personnel. The documents to be submitted are also done on this platform.

→ **The qualification**

The application process is a competitive process that takes place in two distinct phases. First, a period of registration of applications takes place from mid-September to the end of October. Then, a period of transmission of the documents constituting the file takes place from mid-September to mid-December if the thesis is defended before the beginning of November or until mid-January if the thesis is defended between the beginning of November and the beginning of January. The dates vary each year, so it is imperative to consult the calendar.
The general documents required are a copy of your doctoral diploma (or attestation), a copy of your thesis defense report, a CV focusing on your research activities and a copy of your publications (up to three documents). The additional documents to be provided differ according to the CNU section to which you belong. For example, section 60 of the CNU concerns the field of mechanics, mechanical engineering, civil engineering.
In general, to obtain the qualification of MCF in the technical sciences, the candidate must have submitted at least one scientific publication of quality as first author and recognized by the section and must have ensured a minimum number of teaching hours (for example, 64h requested for section 60).
The qualification is valid for four years. If after 4 years you have not obtained a position as an MCF, you will have to reapply for the qualification to continue to compete for the competition for Maître de conférences.

→ **Recruitment**

The list of Lecturer positions to be filled is posted on the Galaxie portal. The recruitment campaign is synchronized at the national level according to a fixed calendar. The registration of applications for the positions and the submission of electronic documents takes place from the end of February to the end of March. Then, in each university and each program, a commission of specialists is made up of teacher-researchers with a mandatory parity. The committee analyses the applications in a first phase, interviews the selected candidates in a second phase, and finally ranks the candidates for the position to be filled. At the end of June, the candidates can consult the decisions of the institutions on the Galaxie portal and express their wishes for an assignment. At the beginning of July, the results are published. If your application is accepted, you become an MCF trainee for the first year before being granted tenure.

6.4 Research Engineer

Research / Teaching: *90 % / 10 %*
Sector : *public*
Duration : *CDI (*Open-ended contract*)*
Salary : *the grid is available* **here**

What is a Research Engineer ?

A research engineer (IR) belongs to the research support professions. He/she has a contractual status and is a category A civil servant assigned to higher education institutions and Public Scientific and Technological Establishments (ESPT).

His/her missions are multiple. He/she participates in the implementation of research activities and the promotion and dissemination of scientific and technical information. He or she directs and coordinates the various technical and administrative activities that contribute to the realization of a research program. He/she may be in charge of all special or general studies and missions. In this capacity, he/she may be entrusted with missions of international cooperation, teaching or research administration. He/she may assume supervisory responsibilities in his/her research unit or department, mainly with regard to technical staff.

How to become a Research Engineer ?

To apply for this position, the candidate must have a PhD or an engineering degree with research experience (internship). The recruitment is done through a competitive process, either at the university or at the ESPT (CNRS, INSERM, INRIA, INRAE, ...).

Concerning the CNRS, the recruitment procedure for IR is as follows. The positions are published at the end of May and beginning of June on the CNRS website. The candidate can apply for several positions at the same time for the Professional Activity Branch (BAP) that interests him/her by giving an order of preference. They must then submit their application file (CV, cover letter, diplomas, etc.) online until the beginning of July. Then, in September, the eligibility phase takes place with the examination of the application files. A jury is appointed for each competition. The selected applications enter the admission phase which takes place until the beginning of November. The candidates are then auditioned: a 10-minute presentation followed by a 20-minute interview. More rarely, a written or practical test may be held. A candidate who is offered an IR position may take up his or her duties on December 1st. He/she becomes an IR trainee for the first year before being granted tenure. Approximately sixty IR positions are awarded each year within the CNRS.

6.5 Chargé de Recherche(CR) (Researcher)

Research / Teaching: *80 % / 20 %*
Sector : *public*
Duration : *CDI (*Open-ended contract*)*

Salary : *the grid is available* here

What is a CR ?

In France, since 1982, the Public Scientific and Technological Establishments (EPST) have had jobs for civil servant researchers. The CNRS, INSERM, INRAE, INRIA, IRD, INED, IRSTEA and IFSTTAR are EPSTs. The researchers of each EPST are divided between the corps of research fellows and the corps of research directors.

A CR is a permanent civil servant classified in category A. He/she is responsible for the development of knowledge, its transfer and application in companies, and in all fields contributing to the progress of society. He disseminates information and scientific and technical culture throughout the population, and particularly among young people. It participates in initial and continuing education and in the administration of research. This position corresponds very closely to that of a university lecturer, with the difference that the researcher has no teaching duties to perform. Although they are not required to teach, they have the possibility of taking on teaching responsibilities, which are paid on an hourly basis.

How to become a CR in CNRS ?

Around November-December, job offers are posted on the CNRS website and online registration is open. At the CNRS, research is divided into sections. For example, section 7 concerns information sciences. It is necessary to find out about the requirements of each section for the preparation of the application file. In particular, a report of about 50 pages mentioning the candidate's previous work and a second report of about 50 pages describing the research project that the candidate would like to implement if admitted to the position are expected. In January, candidates must submit their application, which is analyzed by a jury of approximately 21 members in each section. Those admitted are required to continue with an audition between March and May, consisting of a 12-minute presentation and 15 to 30 minutes of questions. The jury then establishes a list of eligible candidates in order of ranking. In June, the admissions committee selects the candidates who have been admitted and who are then assigned in October. It is very rare to be admitted on your first application. It is a competition that must be repeated several times to refine one's file.

6.6 Other Jobs

If you wish to work abroad, the previous list of professions no longer applies as it is exclusively related to the French civil service. You should then find out about the research and teaching professions in your host country.

It is also possible to go into the private sector after the thesis, and many doctoral students choose this path. The opportunities are varied, you can work in the Research and Development department of a company, in a consulting firm or even set up your own company.

6.7 To see further

The Habilitation à Diriger des Recherches (HDR) is the highest French higher education diploma that can be obtained after a PhD. For the candidate, it provides recognition of his or her scientific level, the original nature of his or her approach in a scientific field, his or her ability to master a research strategy in a scientific or technological field, and his or her capacity to supervise young researchers. The HDR allows its holder to direct theses and to be appointed as thesis rapporteur. It allows the holder to apply for a position as a university professor after being registered on the qualification list by the CNU (similar to the qualification of maître de conférences). The applicant must have obtained his or her doctorate at least 24 months before applying for authorization to register for the HDR.

Within EPST, after a position as a research fellow with at least three years of seniority, it is possible to apply for a senior position as Director of Research (DR). In addition to the tasks assigned to researchers, the RD is responsible for designing, leading or coordinating research or development activities. The competition process is similar to that of a research fellow.

6.8 Sources

https://www.cnrs.fr/fr/page-daccueil

https://www.galaxie.enseignementsup-recherche.gouv.fr/ensup/candidats.html

http://wwwobs.univ-bpclermont.fr/atmos/fr/enseignement/futur_etudiant/apres_these.php

https://www.enseignementsup-recherche.gouv.fr/cid56329/les-personnels-de-la-recherche.html

7 Specific procedures for foreign doctoral students

If you are a foreign doctoral student, there are a certain number of administrative steps to take (visa and immigration, insurance, social security ...). The Euraxess website gathers this information.

8 Support unit

It is common during the PhD to have difficult moments, which can be due to a difficulty in getting to work, the absence of satisfactory results, a block on a technical problem or a bad relationship with the thesis supervisors. If you are confronted with such difficulties, and if they have a clear impact on your mental health, it is important to talk about it around you to defuse the problem as soon as possible. The best people to talk to are your thesis supervisors, but if your relationship with them is strained, it may be difficult to tell them about your situation. In this case, don't hesitate to talk about it with the members of your team with whom you get along well.

If you don't have anyone to talk to, ICUBE has also set up a support unit, which you can contact at mediation@icube.unistra.fr .

There is also a free and anonymous psychological support service at Unistra